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Two electrodes E2/E4 -> four-resistor-per-leg input networks -> dual AMC3330 isolated paths -> ADS1115 -> ESP32.

HI: 997:1 divider; 600 VAC RMS design target. LO: 10.96:1 divider; use conservative +/-2 V working range before characterization.

Both ranges continuously connected. Sensitive path clamps during a fault; test leakage, gain error, saturation and recovery.

External fused 2S harness feeds PCB logic/servos; motor/ESC high-current power stays off PCB. Never parallel ESC BEC.

Remove JP1 before USB. U1 requires WROOM pin availability. Expansion signals are 3.3 V only.

No CAT rating, surge qualification, water-safety threshold or guaranteed detection floor is claimed.

Control

Power

Input_protection

Isolation

ADC

File: Control.kicad_sch

File: Power.kicad_sch

File: Input_protection.kicad_sch

File: Isolation.kicad_sch

File: ADC.kicad_sch

Not qualified for energized-water use; see design and validation record

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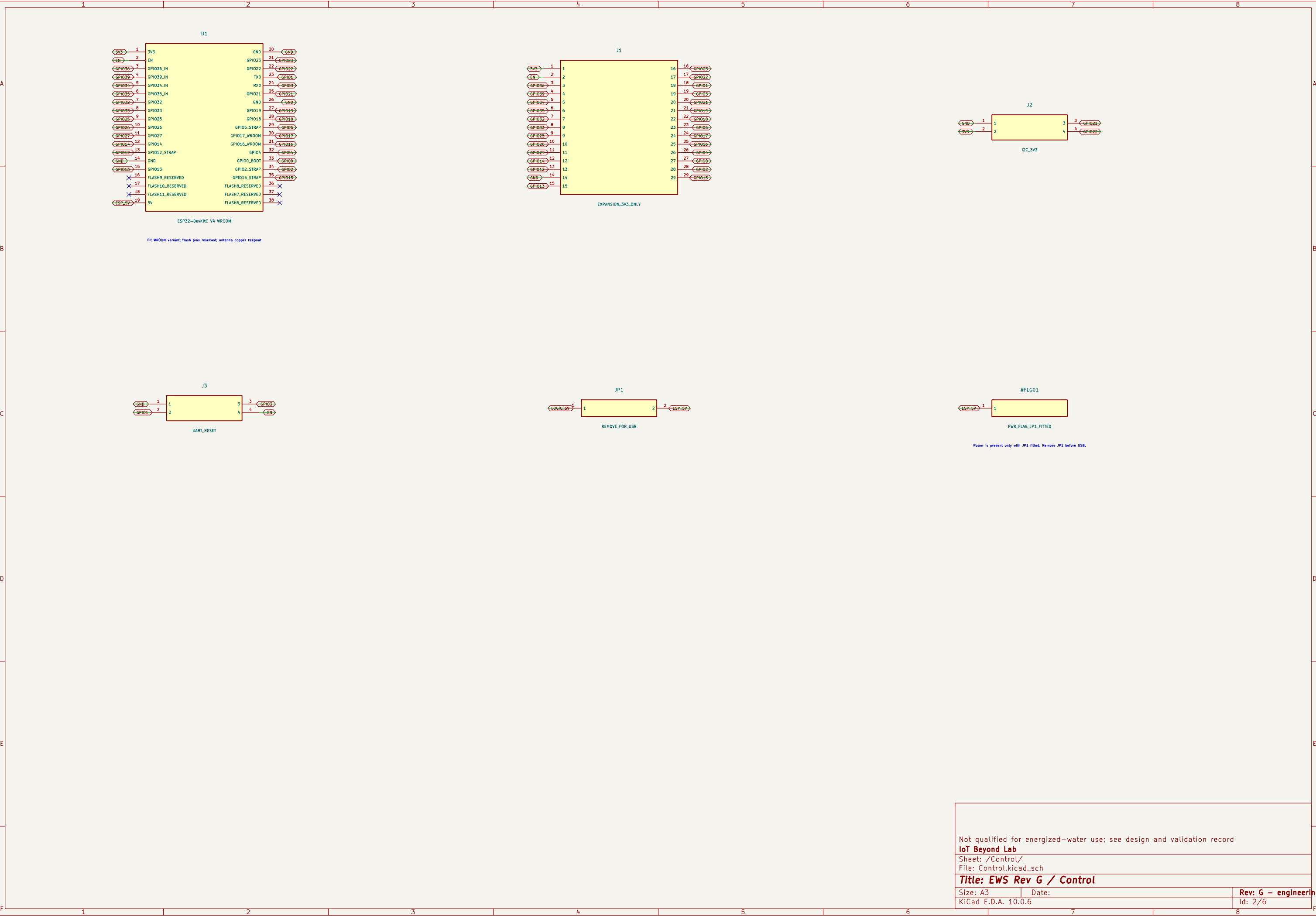
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Title: EWS Rev G / System overview

Size: A3Date:Rev: G - engineering prototype

KiCad E.D.A. 10.0.6Id: 1/6

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Sheet: /Control/

File: Control.kicad_sch

Title: EWS Rev G / Control

Size: A3

Date:

Rev: G - engineering prototype

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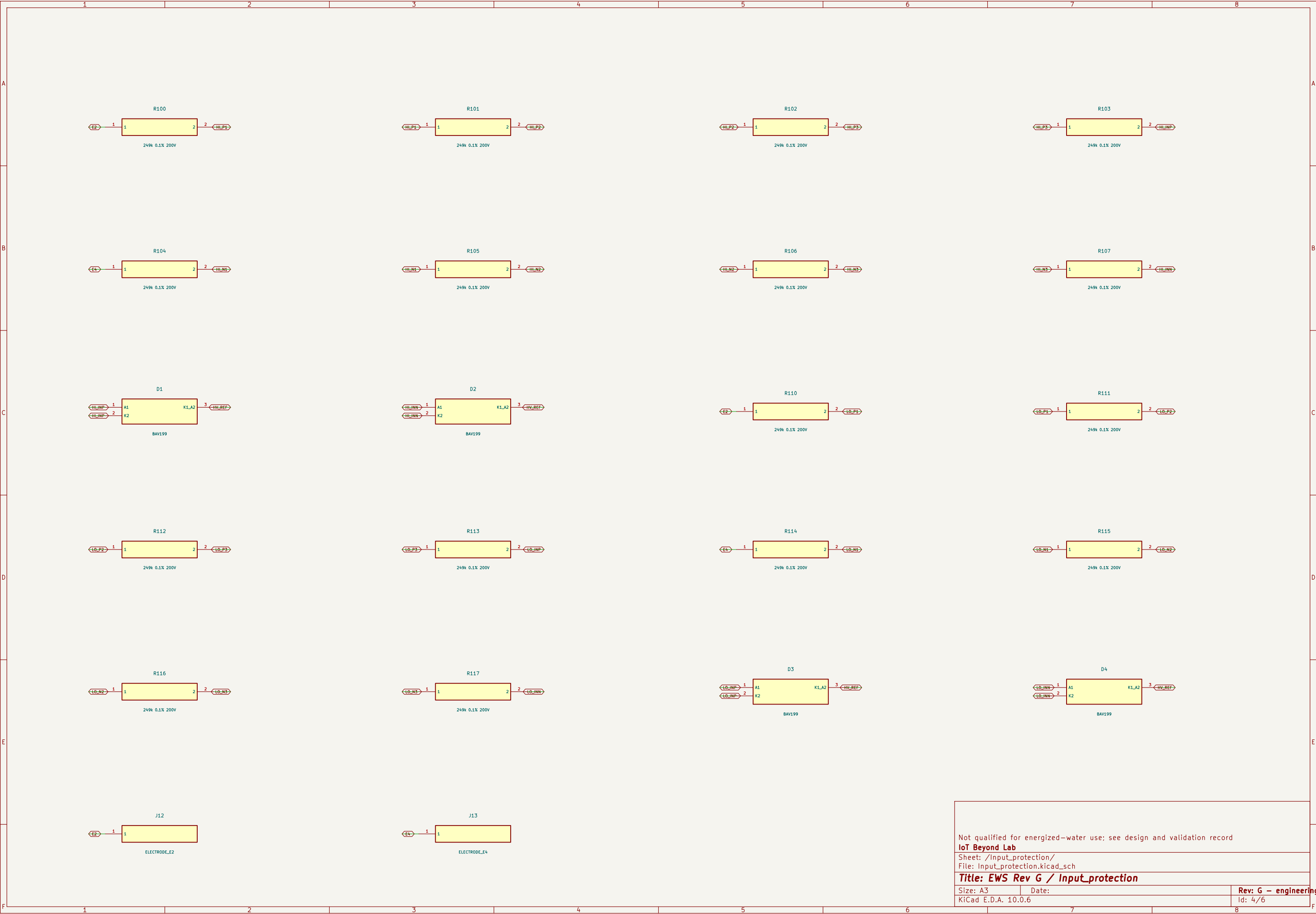
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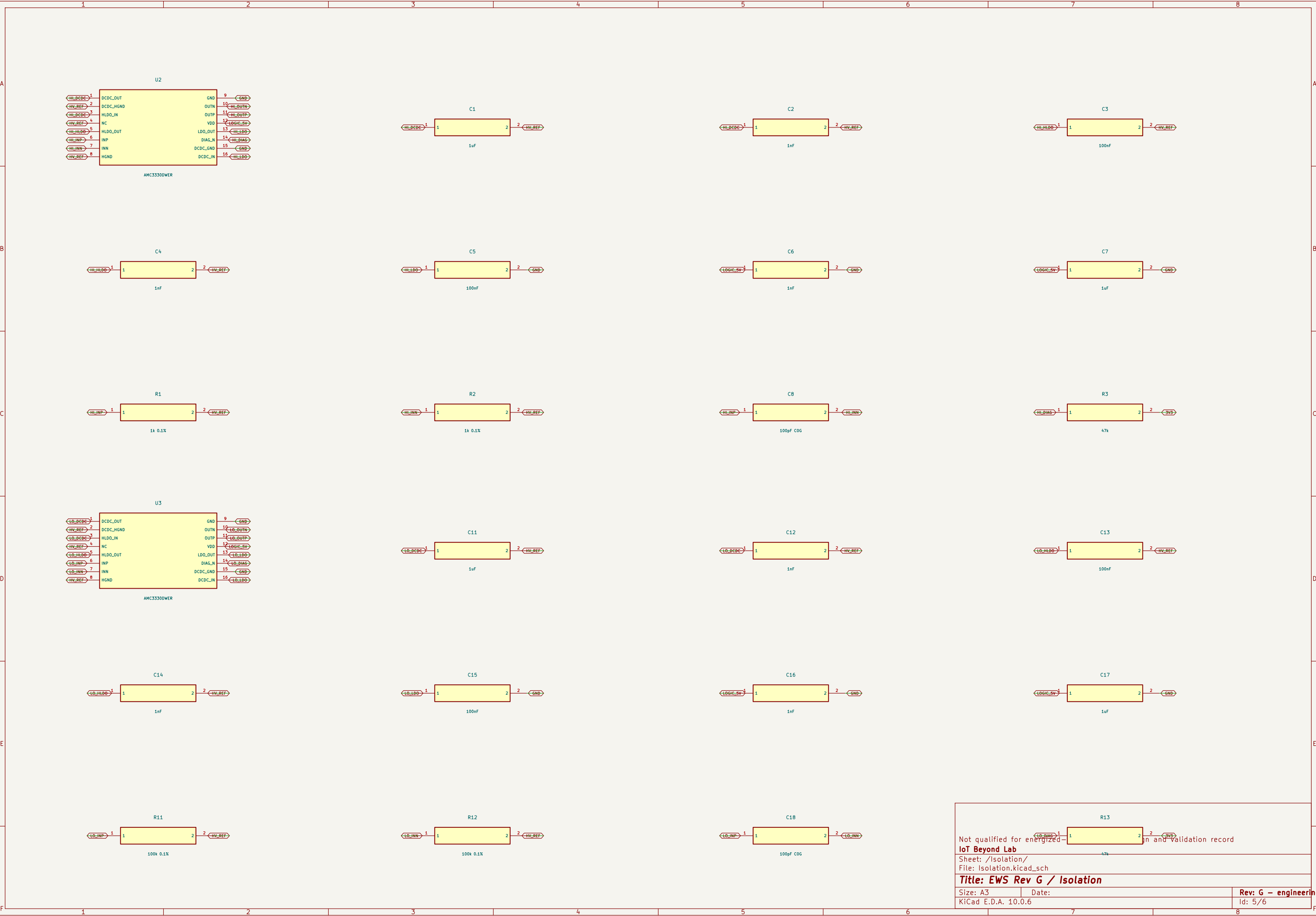


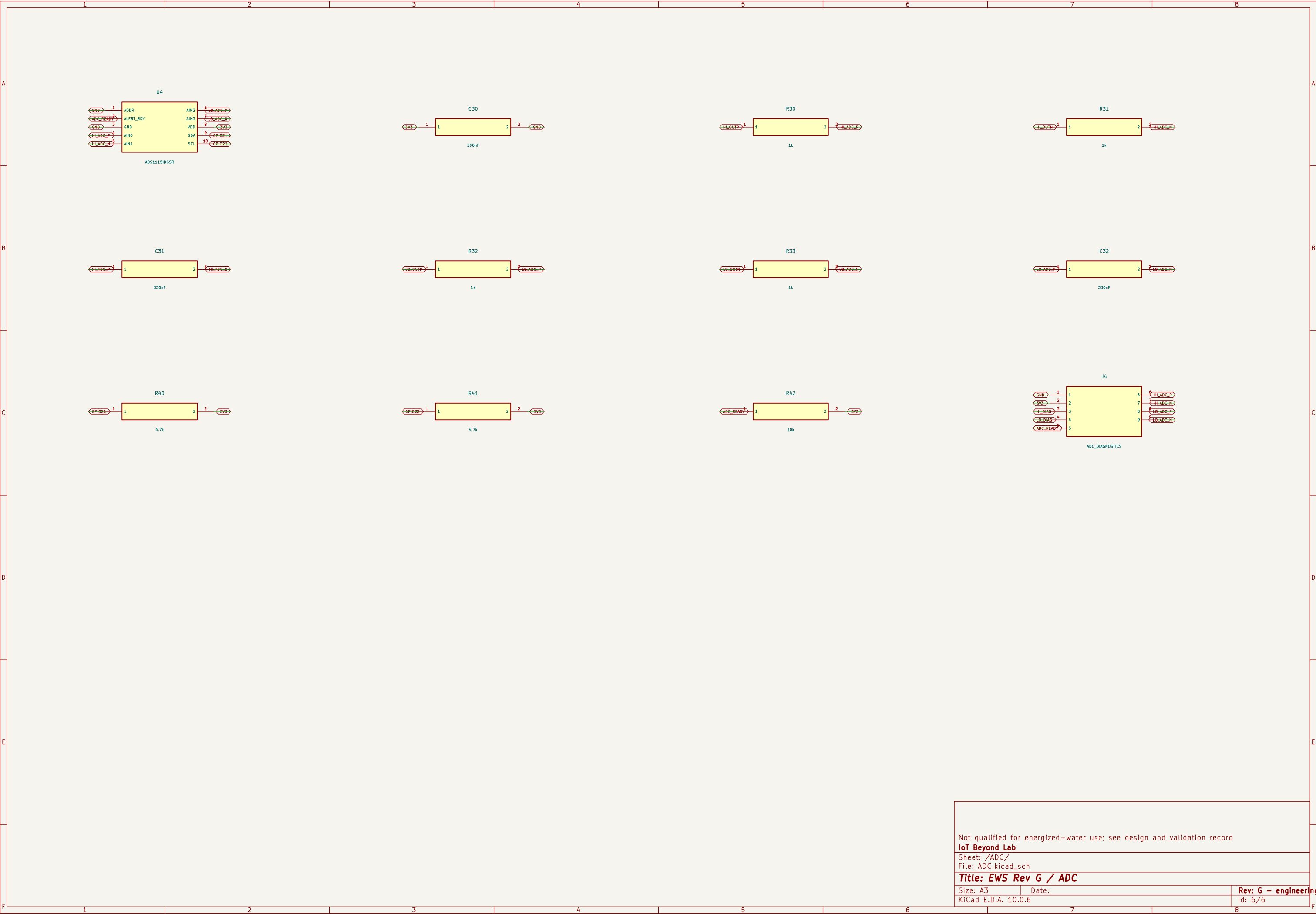
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Title: EWS Rev G / Power

Size: A3	Date:	Rev: G – engineering prototype
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Sheet: /ADC/
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Title: EWS Rev G / ADC

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